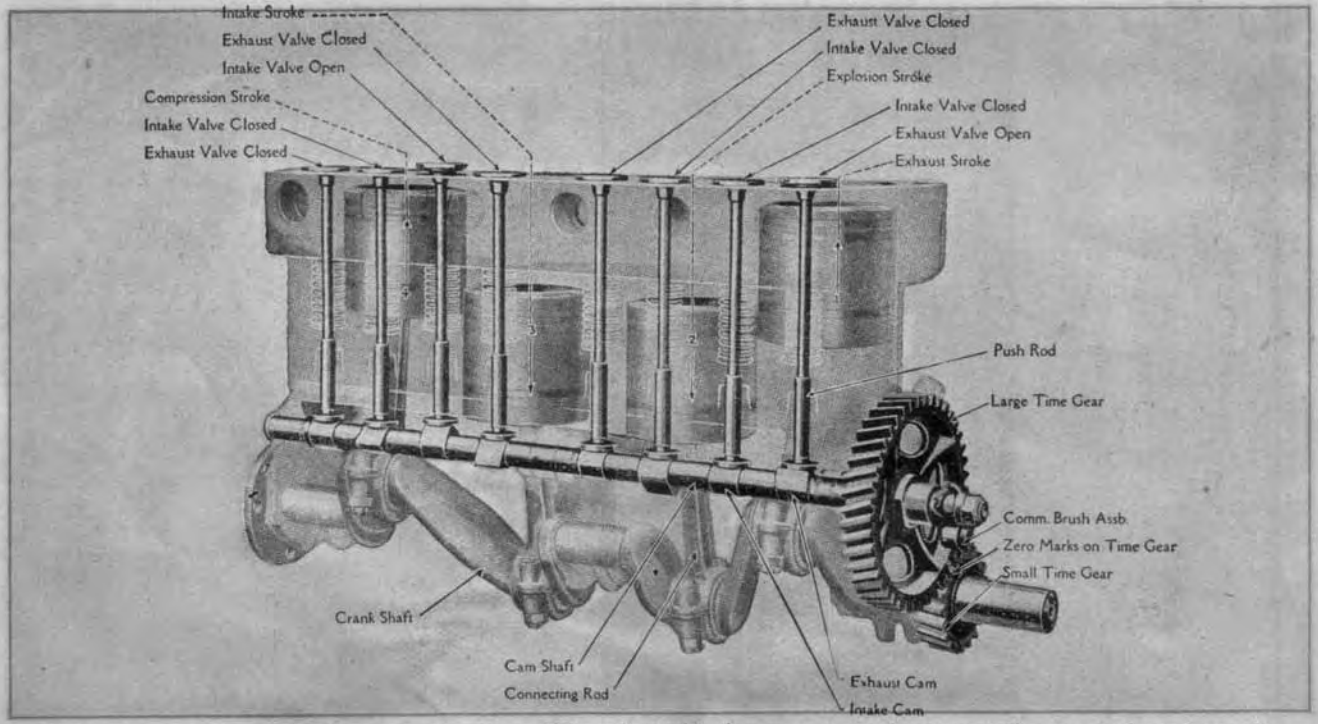
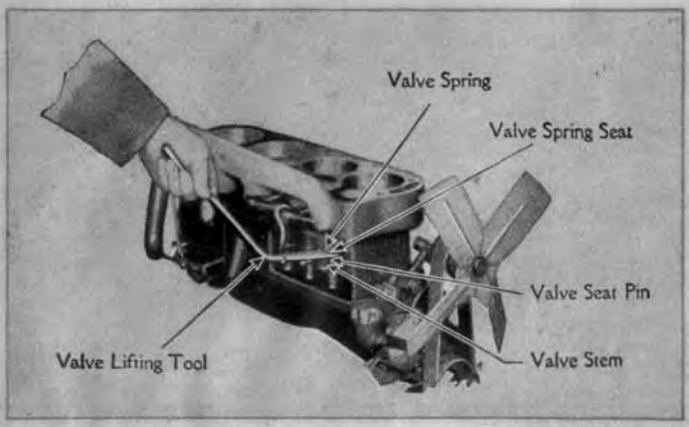


Cylinder Assembly, showing the correct position of the valves with time gears properly set according to punch marks on the gears, also the relative position of the pistons in their strokes as indicated above. The firing order of the cylinders is 1, 2, 4, 3.
(Cut No. 3)

now being properly set, the exhaust valve on No. 1 cylinder is open and the intake valve closed, the other valves being in the position indicated in Cut No. 3. The opening and closing of the valves being as follows: The exhaust valve opens when the piston reaches 5-16″ of bottom center, the distance from the top of the piston head to the top of cylinder casting measuring 3 3-8″. The exhaust valve will close on top centre, the piston being 5-16″ above the cylinder casting. The intake valve opens 1-16″ after top center, and closes 9-16″ after bottom center, the distance from the top of the piston to the top of the cylinder casting measuring 3 1-8″.



How the valve lifting tool should be used. (Cut No. 4)

The clearance between the push rod and valve stem should never be greater than 1-32″ nor less than 1-64″. The correct clearance is naturally half way between these two measurements. The gap should be measured when the push rod is on the heel of the cam.

What about the care of the Valves? — *Answer No. 23*

They seldom get out of order—but they do get dirty, as a result of carbon collecting on the valve seats. These carbon deposits by preventing proper closing of the valves, permit the gases under compression to escape, resulting in loss of power and uneven running of the motor. If, when turning the engine over slowly, there is a lack of resistance in one or more cylinders, it is probable that the valves need re-grinding. As the "life" of the engine depends largely upon the proper seating of the valves, it is necessary that they be ground occasionally.

How are Valves removed for grinding? — *Answer No. 24*

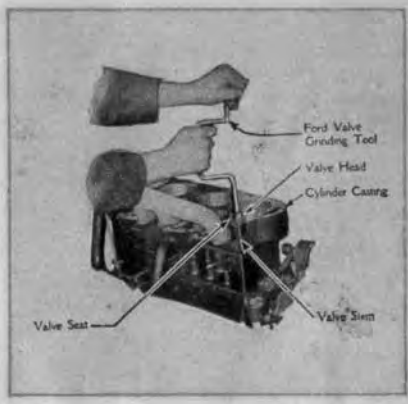
(1) Drain radiator; (2) remove cylinder head; (3) remove the two valve covers on the right side of engine; (4) raise the valve spring with lifting tool (see Cut No. 4) and pull out the little pin under the valve seat. The valve may then be lifted out by the head.

How are Valves ground?

Answer No. 25

For this work use a good grinding paste of ground glass and oil—procurable from auto supply houses. A convenient way is to put a small amount in a suitable dish, adding a spoonful or two of kerosense and a few drops of lubricating oil to make a thin paste.

Place the mixture sparingly on the bevel face of the valve. Put the valve in position on the valve seat, and rotate it back and forth (about a quarter turn) a few times, with a Ford grinding tool. Then lift slightly from the seat, change the position and continue the rotation, and keep on repeating this operation until the bearing surface is smooth and bright. The valve should not be turned through a complete revolution, as this is apt to cause scratches running around the entire circumference of the valve and seat. When the grinding is completed the valve should be removed from the cylinder, thoroughly washed with kerosene, and the valve seat wiped out thoroughly. Extreme care should be taken that no abrasive substance gets into the cylinders or valve guides. This can be avoided if the grinding paste is applied sparingly to the bevel face of the valve. If the valve seat is worn badly or seamed, it is best to have it reseated with a valve seating tool. This operation requires considerable skill, and perhaps had better be done by an expert mechanic. Care should be exercised against making too deep a cut, necessitating the retiming of the valve.



Method of grinding valve. (Cut No. 5)

When the Valves and Push Rods are worn—what?

Answer No. 26

When the valves or push rods become worn, so as to leave too much play between them, thus reducing the lift of the valves and diminishing the power of the motor, it is best to replace the push rods with new ones. The clearance between the push rods and the valve stem should never be greater than $\frac{1}{32}''$ nor less than $\frac{1}{64}''$. If the clearance is greater, the valve will open late and close early, resulting in uneven running of the motor. If the clearance is less than $\frac{1}{64}''$ there is danger of the valve remaining partially open all the time. If replacing the push rod does not give the proper clearance, the valve should also be replaced. We do not recommend drawing out the valve stem, as the operation requires experience and the price of the new part does not warrant the time and expense necessary to properly do the work.

What about the Valve Springs?

Answer No. 27

When the valves fail to seat themselves properly, there is a possibility that the springs may be weak or broken. A weak inlet spring would probably not affect the running of the engine, but weakness in the exhaust valve spring causes a very uneven action, which is difficult to locate. The symptoms are a lag in the engine due to the exhaust valve not closing instantaneously, and as a result a certain percentage of the charge under compression escapes, greatly diminishing the force of the

explosion. Weakness in a valve spring can usually be detected by the following method: Remove the plate which encloses them at the side of the cylinder and insert a screw-driver between the coils of the spring while the engine is running. If the extra tension thus produced causes the engine to pick up speed, the spring is obviously weak and should be replaced with a new one.

What cause "Knocking" in the Engine?

Answer No. 28

There are several causes, which may be enumerated as follows: (1) carbon knock —which is by far the most common—resulting from carbonizing of cylinders; (2) knock caused by a too advanced spark; (3) connecting rod knock; (4) crank shaft main bearing knock; (5) knock due to loose-fitting piston or broken ring; (6) knock caused by the piston striking the cylinder head gasket. When the engine knocks from any cause whatsoever, the matter should be promptly investigated by an experienced mechanic and the difficulty corrected.

How may the different Knocks be distinguished?

Answer No. 29

(1) The carbon knock is a clear, hollow sound, most noticeable in climbing sharp grades, particularly when the engine is heated. It is also indicated by a sharp rap immediately on advancing the throttle. (2) Too advanced spark will be indicated by a dull knock in the motor. (3) The connecting rod knock sounds like the distant tapping of steel with a small hammer, and is readily distinguished when the car is allowed to run idly down grade—or upon speeding the car to twenty-five miles an hour, then suddenly closing the throttle, the tapping will be very distinct. (4) The crank shaft main bearing knock can be distinguished, when the car is going uphill, as a dull thud. (5) The loose piston knock is heard only upon suddenly opening the throttle, when the sound produced might be likened to a rattle. The remedies for these knocks are treated under their proper division.

How is carbon removed from Combustion Chamber?

Answer No. 30

First, drain the water off by opening the pet cock at the bottom of the radiator; then disconnect the wires at the top of the motor and also the radiator connection attached to the radiator. Remove the 15 cap screws which hold the cylinder head in place. Take off the cylinder head and, with a putty knife or screw-driver, scrape the carbonized matter from the cylinder head and top of pistons, being careful to prevent the specks of carbon from getting into the cylinders or bolt holes. In replacing the cylinder head gasket turn the motor over so that No. 1 and No. 4 pistons are at top center; place the gasket in position over the pistons and then put the cylinder head in place. Be sure and draw the cylinder head bolts down evenly (i. e., give each bolt a few turns at a time); do not tighten them on one end before drawing them up at the other.

How are Spark Plugs cleaned?

Answer No. 31

After removing the plug from the engine the points may be cleaned with an old tooth-brush dipped in gasoline. However, in order to do the work thoroughly, the plug should be taken apart by securing the large hexagon steel shell in a vise and loosening the pack nut which holds the porcelain in place. The carbon deposits can then be easily removed from the porcelain and shell with a small knife. Care should be exercised not to scrape off the glazed surface of the porcelain, otherwise it will be apt to carbonize quickly. The porcelain and other parts should finally be washed in gasoline and wiped dry with a cloth.

In assembling the plug care should be taken to see that the pack nut is not tightened too much so as to crack the porcelain, and the distance between the sparking points should be $\frac{1}{32}''$, about the thickness of a smooth dime.

Dirty plugs usually result from an excess of oil being carried in the crank case—or from using oil of poor quality.

How is the Power Plant removed from Car? — *Answer No. 32*

(1) Drain the water out of the radiator and disconnect the radiator hose;

(2) disconnect the radiator stay rod which holds it to the dash;

(3) take out the two bolts which fasten the radiator to the frame and take radiator off;

(4) disconnect the dash at the two supporting brackets which rest on the frame;

(5) loosen the steering post bracket, fastened to the frame, when the dash and steering gear may be removed as one assembly—the wires first having been disconnected;

(6) take out the bolts holding the front radius rods in the socket underneath the crank case;

(7) remove the four bolts at the universal joint;

(8) remove pans on either side of cylinder casting and turn off gasoline, disconnect feed pipe from carburetor;

(9) disconnect exhaust manifold from exhaust pipe by unscrewing large brass pack nut;

(10) take out the two cap screws which hold the crank case to the front frame;

(11) remove the bolts which hold the crank case arms to the frame at the side. Then pass a rope through the opening between the two middle cylinders and tie in a loose knot. Through the rope pass a "2x4," or a stout iron pipe about ten feet long, and let a man hold each end; let a third man take hold of the starting crank handle, when the whole power plant can be lifted from the car to the work bench for adjustment.

Connecting Rod Bearings —how adjusted? — *Answer No. 33*

Connecting rod bearings may be adjusted, without taking out the engine, by the following method: (1) drain off the oil; (2) remove plate on bottom of crank case—exposing connecting rods; (3) take off first connecting rod cap, and draw-file the ends—a very little at a time; (4) replace cap, being careful to see that file marks correspond, and tighten bolts until it fits shaft snugly; (5) test tightness of bearing by turning engine over by the starting handle. Experienced mechanics usually determine when the bearing is properly fitted by lightly tapping each side of the cap with a hammer; (6) then loosen the bearing and proceed to fit the other bearings in the same manner; (7) after each bearing has been properly fitted and tested—then tighten the cap bolts and the work is finished.

Remember, there is a possibility of getting the bearings too tight, and under such conditions the babbitt is apt to cut out quickly, unless precaution is taken to run the motor slowly at the start. It is a good plan after adjusting the bearings to jack up the rear wheels and let the motor run slowly for about two hours (keeping it well supplied with water and oil) before taking it out on the road. Whenever possible these bearings should be fitted by an expert Ford mechanic.

Worn connecting rods may be returned, prepaid, to the nearest dealer or branch house for exchange at a price of $1.00 each to cover the cost of rebabbitting. It is not advisable for any owner or repair shop to attempt the rebabbitting of connecting rods or main bearings, for without a special jig in which to form the bearings,

satisfactory results will not be obtained. The constant tapping of a loose connecting rod on the crank shaft will eventually produce crystallization of the steel—result, broken crank shaft and possibly other parts of the engine damaged.

Crank Shaft Main Bearings —how adjusted?

Answer No. 34

Should the stationary bearings in which the crank shaft revolves become worn (evidenced by a pounding in the motor) and need replacing or adjusting, proceed as follows:

(1) After the engine has been taken out of the car, remove crank case, transmission cover, cylinder head, pistons, connecting rods, transmission and magneto coils. Take off the three babbitted caps and clean the bearing surfaces with gasoline. Apply Prussian blue or red lead to the crank shaft bearing surfaces, which will enable you, in fitting the caps, to determine whether a perfect bearing surface is obtained.

(2) Place the rear cap in position and tighten it up as much as possible without stripping the bolt threads. When the bearing has been properly fitted, the crank shaft can be turned with one hand. If the crank shaft cannot be turned with one hand, the contact between the bearing surfaces is evidently too close, and the cap requires shimming up, one or two brass liners usually being sufficient. In case the crank shaft moves too easily with one hand, the shims should be removed and the steel surface of the cap filed off, permitting it to set closer.

(3) After removing the cap, observe whether the blue or red "spottings" indicate a full bearing the length of the cap. If "spottings" do not show a true bearing, the babbitt should be scraped and the cap refitted until the proper results are obtained.

(4) Lay the rear cap aside and proceed to adjust the center bearing in the same manner. Repeat the operation with the front bearing, with the other two bearings laid aside.

(5) When the proper adjustment of each bearing has been obtained, clean the babbitt surface carefully and place a little lubricating oil on the bearings, also on the crank shaft; then draw the caps up as closely as possible—the necessary shims, of course, being in place. Do not be afraid of getting the cap bolts too tight, as the shim under the cap and the oil between the bearing surfaces will prevent the metal being drawn into too close contact. If oil is not put on the bearing surfaces, the babbitt is apt to cut out when the motor is started up before the oil in the crank case can get into the bearing. In replacing the crank case and transmission cover on the motor, it is advisable to use a new set of gaskets to prevent oil leaks.

The Ford Cooling System

How is the Engine cooled? — *Answer No. 35*

The heat generated by the constant explosions in the engine would soon overheat and ruin the engine, were it not cooled by some artificial means. The Ford engine is cooled by the circulation of water in jackets around the cylinders. The heat is extracted from the water by its passing through the thin metal tubing of the radiator—to which are attached scientifically worked out fins, which assist in the rapid radiation of the heat. The fan, just back of the radiator, sucks the air around the tubing—around which the air is also driven by the forward movement of the car. The belt should be inspected frequently and tightened when necessary—not too tight, however—by means of the adjusting screw in the fan bracket. Take up the slack till the fan starts to bind when turned by hand.

How does the Water circulate? — *Answer No. 36*

The cooling apparatus of the Ford car is known as the Thermo-syphon system. It acts on the principle that hot water seeks a higher level than cold water—consequently when the water reaches a certain heat, approximately 180 degrees Fahrenheit, circulation commences and the water flows from the lower radiator outlet pipe up through the water jackets, into the upper radiator water tank, and down through the tubes to the lower tank, to repeat the process.

What are the causes of Overheating? — *Answer No. 37*

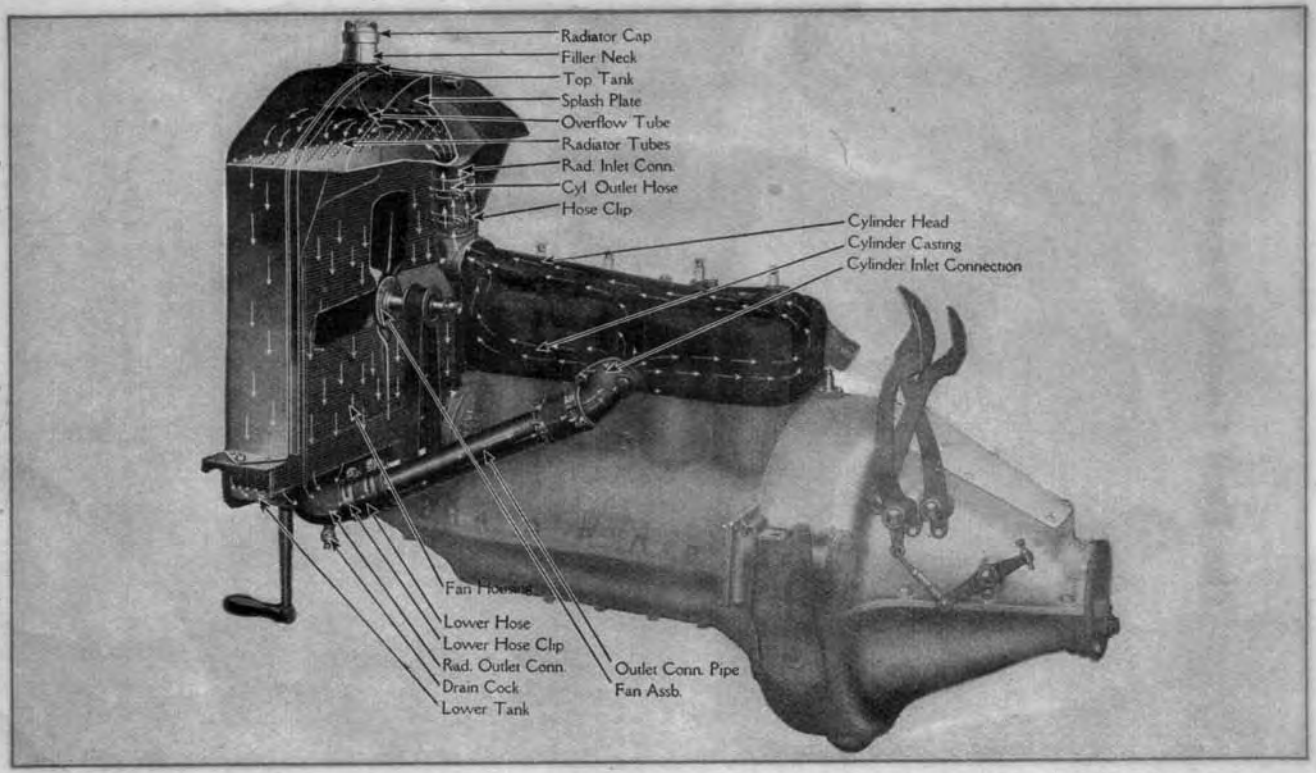
(1) Carbonized cylinders; (2) too much driving on low speed; (3) spark retarded too far; (4) poor ignition; (5) not enough or poor grade oil; (6) racing motor; (7) clogged muffler; (8) improper carburetor adjustment; (9) fan not working properly on account of broken or slipping belt; (10) improper circulation of water due to clogged or jammed radiator tubes, leaky connections or low water.

What should be done when the Radiator overheats? — *Answer No. 38*

Keep the radiator full. Don't get alarmed if it boils occasionally—especially in driving through mud and deep sand or up long hills in extremely warm weather. Remember that the engine develops the greatest efficiency when the water is heated nearly to the boiling point. But if there is persistent overheating when the motor is working under ordinary conditions—find the cause of the trouble and remedy it. The chances are that the difficulty lies in improper driving or carbonized cylinders. Perhaps twisting the fan blades at a greater angle to produce more suction may bring desired results. By reference to the proper division of this book each of the causes which contribute to an overheated radiator is treated and remedies suggested. No trouble can result from the filling of a heated radiator with cold water—providing the water system is not entirely empty—in which case the motor should be allowed to cool before the cold water is introduced.

How about cleaning the Radiator? — *Answer No. 39*

The entire circulating system should be thoroughly flushed out occasionally. To do this properly, the radiator inlet and outlet hose should be disconnected, and the radiator flushed out by allowing the water to enter the filler neck at ordinary pressure, from whence it will flow down through the tubes and out at the drain cock and hose. The water jackets can be flushed out in the same manner. Simply



The Thermo-Syphon Cooling System, showing course of water through water passages. (Cut No. 6)

allow the water to enter into the cylinder head connection and to flow through the water jackets and out at the side inlet connection.

Will the Radiator freeze in winter?

Answer No. 40

Yes, unless an anti-freezing solution is used in the circulating system you are bound to experience trouble. As the circulation does not commence until the water becomes heated, it is apt to freeze at low temperature before it commences to circulate. In case any of the radiator tubes happen to be plugged or jammed they are bound to freeze and burst open if the driver undertakes to get along without using a non-freezing solution. Wood or denatured alcohol can be used to good advantage. The following table gives the freezing points of solutions containing different percentages of alcohol:

20% solution freezes at 15 degrees above zero.
30% solution freezes at 8 degrees below zero.
50% solution freezes at 34 degrees below zero.

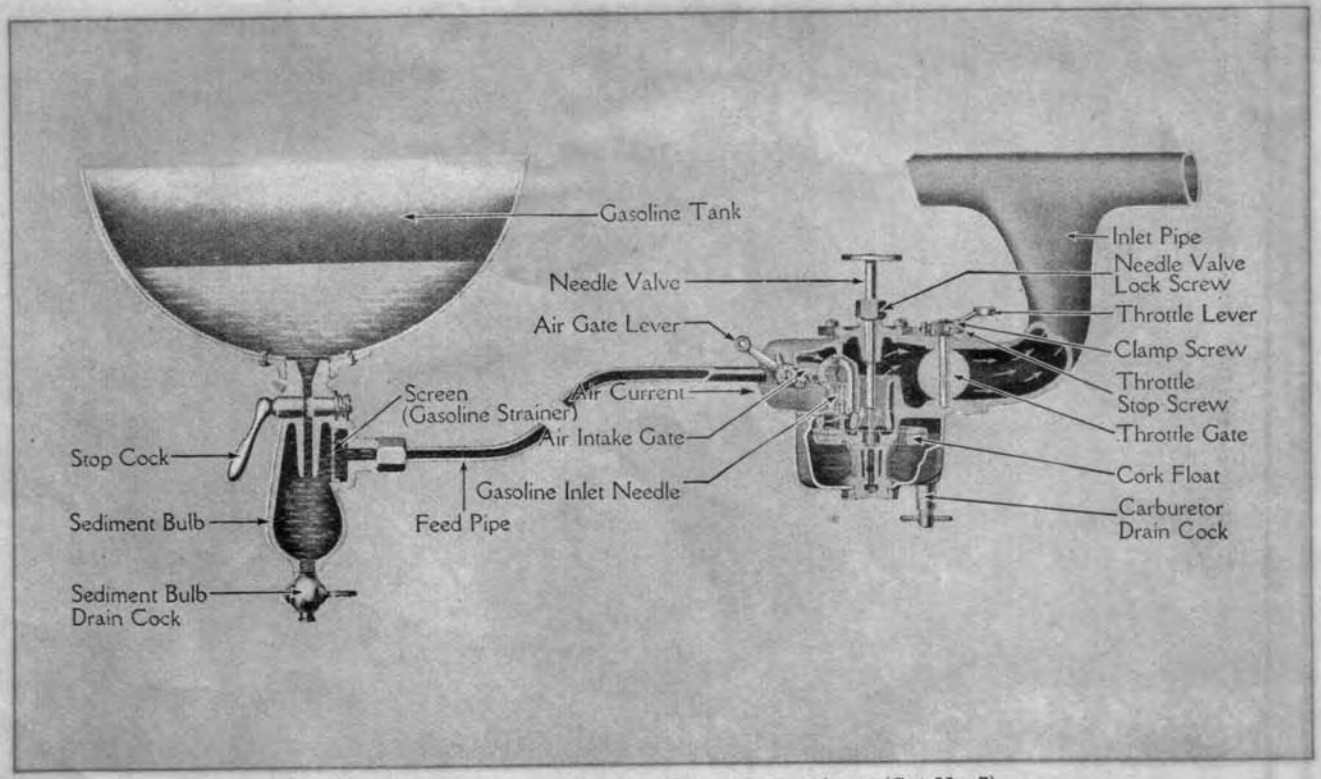
A solution composed of 60% water, 10% glycerine and 30% alcohol is commonly used, its freezing point being about 8 degrees below zero.

On account of evaporation fresh alcohol must be added frequently in order to maintain the proper solution.

How are leaks and jams in the Radiator repaired?

Answer No. 41

A small leak may be temporarily repaired by applying brown soap or white lead—but the repair should be made permanent with solder as soon as possible. A jammed radiator tube is a more serious affair. While the stopping of one tube does not seriously interfere with the circulation, it is bound to cause trouble sooner or later—and the tube will freeze in cold weather. Cut the tube an inch above and below the jam and insert a new piece, soldering the connections. If the entire radiator is badly jammed or broken it would probably be advisable to install a new one.



This cut illustrates the principle of Ford Carburetion. (Cut No. 7)